

CS498 Week 03-06

February 2026

Outline

I advise completing this section and starting the next **ASAP**. The foundations must be covered before we can start doing more advanced walkthrough.

Shift from “pretest & paperwork” into hands on practice. Students learn how web apps work at a protocol level (HTTP, cookies/sessions, DNS/TCP/IP) and then use Burp Suite as their primary tool for inspecting and manipulating web traffic.

Web Foundations

This section focuses on understanding modern web and networking fundamentals: how requests move through a network, how HTTP is structured, where state lives (cookies/sessions), and what “common web attack vectors” look like in practice. This is the conceptual base students need before Burp Suite makes sense.

Material

A * indicates material for people who haven’t taken Computer Networks. If you have taken computer networks, I would still recommend brushing through the material and understanding the key points.

- [TryHackMe: Web Hacking Fundamentals](#)
 - Overlapping material in this module & assigned module. I recommend completing this after the assigned module.
- *[TryHackMe: How Websites Work](#)
- *[TryHackMe: HTTP in Detail](#)
- *[TryHackMe: Web Application Basics](#)

- [MDN: HTTP messages \(request/response structure\)](#)
- [Cloudflare: OSI Model](#)
- [IBM: Network topology](#)
- [OWASP Top 10 \(common web risks\)](#)

Burp Suite

Burp Suite is the key tool for web testing. Students will learn how to configure a proxy, intercept traffic, inspect and modify requests/responses, and replay requests.

Material

- [PortSwigger: Burp Suite documentation](#)
- [PortSwigger: Intercepting HTTP traffic with Burp Proxy](#)
- [PortSwigger: Modifying requests in Burp Proxy](#)
- [PortSwigger: Burp Proxy](#)
- [TryHackMe: OWASP Juice Shop \(room; applied Burp practice\)](#)
- [YouTube video: OWASP Juice Shop \(Walkthrough\)](#)

Optional Material

If you complete this section early, I advise attempting the following. It will prepare you for the capstone. If you get stuck, reference the tutorial. Comprehend what is happening and where this step fits into the overall penetration testing cycle.

After completing, fill out your LaTeX formatted report. There are example PDFs for each walkthrough tutorial. Try not to use these until the end. I am fine with copying the trivial information (Confidentiality Statement, Contact Info, Assesment Overview), however, fill out the important information yourself to prepare for industry standards. This is the Scope, Executive Summary, Findings Severity Ratings, Security Strengths, Security Weaknesses, and remediation.

Use the LaTeX formatted report from Week 1. Conduct recon, find vulnerabilities, reasearch CVEs, exploit, collect evidence, justify the CVSS score evidence. Don't copy from the tutorial, try to find the vulnerability yourself. Use the appendix section to reference large outputs of evidence. Create clear remediation steps, and an executive summary. Once complete send the document through Discord. If you have any questions please reach out.

- [TryHackMe: Kenobi room \(Look Ahead to Week 12\)](#)

- [YouTube: Kenobi room walkthrough](#)
- [TryHackMe: RootMe room \(Look Ahead to Week 12\)](#)
- [YouTube: RootMe room walkthrough](#)

Deliverables

- Google Form for Quiz 3, submit through Discord
- [Complete Sections 1-4 on TryHackMe's "Jr Penetration Tester" Path](#)
- [TryHackMe: Network Services room \(Start\)](#)
 - [YouTube: Network Services room walkthrough](#)

Quiz 3 Topics

- HTTP request/response anatomy (methods, headers, cookies, status codes)
- Sessions/auth basics (cookies vs tokens, CSRF concept)
- Common web vulns at a high level (IDOR, XSS, SQLi, SSRF, file upload risk)
- What Burp does & where it fits in testing (proxy, intercept, repeater concept)